

Revised Generation Interconnection System Impact Study Report

For Queue Project Q16-0577

Yeagertown 138 kV
December 26, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0577. The Interconnection Customer proposes a 97.3 MW solar photovoltaic facility interconnecting at the Yeagertown 138 kV substation in the Pennsylvania area. The request was received on January 20, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 313 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 17 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$124,446,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Walnut Creek-Volcano 138 KV is overloaded to 254.2% of its 240 MVA emergency rating for the loss of the Coronet-Ft Humbug 500 KV Line. The project contributes 12.3% DFAX (12.0 MW) to this overload.

The Midwind-High Sheldon 138 KV is overloaded to 188.1% of its 151 MVA emergency rating for the loss of the Walnut Creek-Volcano 138 KV. The project contributes 5.9% DFAX (5.7 MW) to this overload.

Buxton-Northside Drive 138 KV: 174.3% loading (149 MVA) following the outage of the Coronet-Fort Humbug 500 KV; DFAX = 18.6%.

For the loss of Ft Humbug 345/138 kV XFMR, the Southbase-Kings 138 KV loads to 161.6% of its 242 MVA rating. Project DFAX: 6.1%.

Milwaukee-Southbase 138 KV: 149.1% loading (241 MVA) following the outage of the Fort Humbug 345/138kV XFMR; DFAX = 8.8%.

The Silverton-Midwind 138 KV Line is overloaded to 148.6% of its 199 MVA emergency rating for the loss of the Volcano-Walnut Crk 138 KV. The project contributes 5.9% DFAX (5.7 MW) to this overload.

For the loss of Ft Humbug Substation-Coronet 500 KV, the Northside Drive Substation-Volcano 138 KV line loads to 142.3% of its 180 MVA rating. Project DFAX: 18.6%.

For the loss of Walnut Crk-Volcano 138 KV, the Bass Lake-High Sheldon 138 KV loads to 140.9% of its 195 MVA rating. Project DFAX: 5.9%.

Fort Humbug 345 kV/138 kV Auto: 133.7% loading (356 MVA) following the outage of the W Campus-Fort Humbug 345 KV; DFAX = 48.5%.

Bass Lake Substation-Hernando 138 KV: 132.6% loading (204 MVA) following the outage of the Walnut Creek-Volcano 138 KV; DFAX = 5.9%.

Sunbury 500 kV/345 kV XFMR: 131.2% loading (976 MVA) following the outage of the Coronet-Westwood Generation Llc Substation 500 KV; DFAX = 7.9%.

The Buxton Substation-Yeagertown Substation 138 KV Line is overloaded to 129.1% of its 206 MVA emergency rating for the loss of the Coronet-Fort Humbug 500 KV. The project contributes 18.6% DFAX (18.1 MW) to this

overload.

For the loss of Fort Humbug Substation-Little Chute 345 KV, the Hastings-Elk City Llc 345 KV loads to 124.6% of its 1,005 MVA rating. Project DFAX: 6.0%.

Walnut Crk-Hernando 138 KV Line: 124.2% loading (212 MVA) following the outage of the Walnut Crk-Volano 138 KV line; DFAX = 5.9%.

Ft Humbug 500/345kV Autotransformer: 107.5% loading (1,409 MVA) following the outage of the Orchrd Rd-Ft Humbug 500 KV; DFAX = 16.4%.

For the loss of Bixby Hill-Brunner Island 500 KV, the Fayette-Ironwood Lc 345 KV L loads to 106.5% of its 1,443 MVA rating. Project DFAX: 5.2%.

Coronet 500/345 kV Transformer: 102.8% loading (739 MVA) following the outage of the Coronet-Ft Humbug 500 KV; DFAX = 6.2%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Walnut Creek-Volcano 138 KV	Rebuild line with higher-capacity conductor	\$5,664,000	100% (\$5,664,000)
Midwind-High Sheldon 138 KV	Rebuild line with higher-capacity conductor	\$5,345,000	100% (\$5,345,000)
Buxton-Northside Drive 138 KV	Rebuild line with higher-capacity conductor	\$5,154,000	100% (\$5,154,000)
Southbase-Kings Substation 138 KV	Rebuild line with higher-capacity conductor	\$10,369,000	100% (\$10,369,000)
Milwaukee-Southbase 138 KV L	Rebuild line with higher-capacity conductor	\$8,663,000	100% (\$8,663,000)
Silverton-Midwind 138 KV Line	Rebuild line with higher-capacity conductor	\$11,449,000	100% (\$11,449,000)
Northside Drive-Volcano Substation 138 KV	Rebuild line with higher-capacity conductor	\$6,184,000	100% (\$6,184,000)
Bass Lake-High Sheldon 138 KV	Rebuild line with higher-capacity conductor	\$9,958,000	100% (\$9,958,000)
Fort Humbug 345 kV/138 kV Auto	Replace transformer with higher-rated unit	\$27,194,000	100% (\$27,194,000)
Bass Lake Substation-Hernando 138 KV	Rebuild line with higher-capacity conductor	\$7,143,000	100% (\$7,143,000)
Sunbury 500/345 kV TX	Replace transformer with higher-rated unit	\$17,090,000	11% (\$1,810,000)
Buxton Substation-Yeagertown Substation 138 KV Line	Rebuild line with higher-capacity conductor	\$9,341,000	100% (\$9,341,000)
Hastings-Elk City Llc 345 KV	Rebuild line with higher-capacity conductor	\$21,787,000	3% (\$643,000)
Walut Creek-Hernando 138 KV	Rebuild line with higher-capacity conductor	\$7,312,000	100% (\$7,312,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Fort Humbug 500 kV/345 kV Autotransformer	Replace transformer with higher-rated unit	\$22,305,000	8% (\$1,663,000)
Fayette-Ironwood Llc 345 KV	Reconductor line	\$9,333,000	2% (\$148,000)
Coronet 500/345 kV Transformer	Replace transformer with higher-rated unit	\$29,108,000	22% (\$6,406,000)
	Total allocated to project		\$124,446,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.